

Subquadratic p -Elastic Flow Near Nonzero Curvature: An $H^4 \rightarrow L^2$ Convergence Extension

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Status and source question

Status: candidate_partial_likely_valid.

Pozzetta's *Convergence of elastic flows of curves into manifolds*, arXiv:2007.00582 [1], studies

$$\mathcal{E}_p(\gamma) = \int_{\mathbb{S}^1} \left(1 + \frac{1}{p}|k|^p\right) ds$$

and its geometric fourth-order flow for $p \geq 2$. On PDF page 5 the paper states that understanding the behavior for $p \in [1, 2)$ remains open. This packet proves a conditional convergence theorem for the full subquadratic range $1 < p < 2$ whenever the curvature stays away from zero. It also explains why this does not yet settle the broad source problem.

Let $p' = p/(p-1)$ and write the normal gradient as

$$G_p(\gamma) = \nabla_s^2(|k|^{p-2}k) + \frac{1}{p'}|k|^p k - k + R(|k|^{p-2}k, \tau)\tau. \quad (1)$$

The flow is $\partial_t \gamma = -G_p(\gamma)$, up to tangential reparametrization.

New result

Theorem 1 (subquadratic nonzero-curvature regime). *Let $1 < p < 2$, and let (M, g) be a complete real-analytic Riemannian manifold with real-analytic metric.*

- (i) *If a smooth immersed closed curve γ_0 satisfies $\min_{\mathbb{S}^1} |k_{\gamma_0}| > 0$, then (1) has a unique smooth short-time solution, modulo reparametrization, and its curvature remains nonzero for a short time.*
- (ii) *If γ_* is a smooth critical point of $\mathcal{E}_p$ with $\min_{\mathbb{S}^1} |k_{\gamma_*}| > 0$, there are $C, \rho > 0$ and $\theta \in (0, 1/2]$ such that every H^4 normal graph γ_v over γ_* with $\|v\|_{H^4} < \rho$ satisfies*

$$|\mathcal{E}_p(\gamma_v) - \mathcal{E}_p(\gamma_*)|^{1-\theta} \leq C \|G_p(\gamma_v)\|_{L^2(ds_{\gamma_v})}. \quad (2)$$

- (iii) *Let $\gamma : [0, \infty) \times \mathbb{S}^1 \rightarrow M$ be a global smooth solution with $|k_\gamma| > 0$. Assume that there are times $t_j \rightarrow \infty$, ambient isometries I_j , reparametrizations ϕ_j , and a smooth critical curve γ_* with $\min |k_{\gamma_*}| > 0$ such that*

$$I_j \circ \gamma(t_j, \phi_j(\cdot)) \longrightarrow \gamma_* \quad \text{in } C^m(\mathbb{S}^1) \quad \text{for every } m.$$

Then the flow converges smoothly to a critical curve as $t \rightarrow \infty$, up to reparametrization. Thus the source theorem's "sub-convergence implies full convergence" conclusion extends to $1 < p < 2$ in the nonzero-curvature regime.

The exponent obstruction and the Hilbert repair

The source realizes the first variation in $L^{p'}$ against L^p and, in the convergence argument, uses

$$\|V\|_{L^2} \geq c\|V\|_{L^{p'}}.$$

This is available on a finite-length curve when $p' \leq 2$, equivalently $p \geq 2$. For $1 < p < 2$ one has $p' > 2$ and the inclusion reverses. Moreover, an $H^{4,p}$ variation naturally places the fourth-order expression in L^p , not in $L^{p'}$. Merely changing the stated range of p in the source proof therefore does not work.

The repair is to use normal fields in the Hilbert pair

$$V = H^4(\mathbb{S}^1; N\gamma_*), \quad Z = L^2(\mathbb{S}^1; N\gamma_*),$$

and represent the first variation by its fixed-density L^2 gradient. This produces (2) directly in precisely the norm occurring in energy dissipation.

Proof of the local analytic and elliptic statements

Put $e = k/|k|$ and define, on the normal bundle,

$$A_k = |k|^{p-2}(I + (p-2)e \otimes e). \quad (3)$$

For a normal vector $\xi = \xi_\perp + \xi_\parallel$, with $\xi_\parallel$ parallel to k , direct calculation gives

$$\langle A_k \xi, \xi \rangle = |k|^{p-2} (|\xi_\perp|^2 + (p-1)|\xi_\parallel|^2). \quad (4)$$

Thus A_k is positive definite for every $p > 1$. On an annulus $0 < c \leq |k| \leq C_0$, its ellipticity constants are uniform. Notice that the possibly negative coefficient $p-2$ in (3) causes no problem; the longitudinal eigenvalue is $(p-1)|k|^{p-2} > 0$.

Choose a normal-graph chart $v \mapsto \gamma_v = \exp_{\gamma_*} v$. Since $H^4(\mathbb{S}^1)$ embeds into C^3 , small H^4 graphs remain immersed and obey $|k_{\gamma_v}| \geq c/2$. Curvature is an analytic function of the graph and its first two derivatives. The scalar function $r \mapsto r^{p/2}$ is real analytic for $r > 0$, so both

$$v \mapsto \mathcal{E}_p(\gamma_v) \in \mathbb{R}, \quad v \mapsto \mathcal{M}(v) \in L^2$$

are analytic; here $\mathcal{M}(v)$ is (1), transported to $N\gamma_*$ and multiplied by the analytic density change. The second variation at the critical point has the form

$$Lv = \nabla_s^2(A_k \nabla_s^2 v) + Kv, \quad L : H^4 \longrightarrow L^2, \quad (5)$$

where K contains at most three derivatives of v . Hence $K : H^4 \rightarrow L^2$ is compact by Rellich's theorem.

Let $P = \nabla_s^2(A_k \nabla_s^2)$. Equations (4) and standard one-dimensional interpolation give the coercive estimate

$$\|v\|_{H^2}^2 \leq C \left(\int_{\mathbb{S}^1} \langle A_k \nabla_s^2 v, \nabla_s^2 v \rangle ds + \|v\|_{L^2}^2 \right).$$

Lax–Milgram followed by fourth-order elliptic regularity shows that $P + I : H^4 \rightarrow L^2$ is an isomorphism. Therefore P , and then $L = P + K$, is Fredholm of index zero. Chill's analytic gradient theorem [2, Corollary 3.11], in the $H^4 \rightarrow L^2$ realization, now gives (2). This proves part (ii).

For part (i), fix a standard tangential gauge. In a local chart the fourth-order principal symbol is, up to a positive power of $|\partial_x \gamma|$, $A_k \xi^4$. Formula (4) makes the gauged system uniformly parabolic at γ_0 . Classical quasilinear fourth-order parabolic theory gives a unique smooth gauged solution. The usual geometric-uniqueness argument removes the gauge, and continuity keeps $|k|$ positive for a short time.

Proof of conditional full convergence

Only the exponent-sensitive part of the source's convergence argument needs replacement. We give the trapping mechanism explicitly.

After choosing a sufficiently late subsequence time, applying its isometry, and using a constant-speed parametrization, start the flow at a curve ε -close to γ_* in every fixed C^m norm. Let T be the maximal time for which the reparametrized flow remains in the normal-graph H^4 ball where (2) holds and $|k| \geq \frac{1}{2} \min |k_{\gamma_*}|$. On $[0, T)$ set

$$H(t) = (\mathcal{E}_p(\gamma(t)) - \mathcal{E}_p(\gamma_*))^\theta.$$

The normal flow satisfies the exact dissipation identity

$$-\frac{d}{dt}\mathcal{E}_p(\gamma(t)) = \|G_p(\gamma(t))\|_{L^2(ds)}^2. \quad (6)$$

Combining (2) and (6) yields

$$-H'(t) = \theta \frac{\|G_p(\gamma(t))\|_2^2}{(\mathcal{E}_p(\gamma(t)) - \mathcal{E}_p(\gamma_*))^{1-\theta}} \geq c \|G_p(\gamma(t))\|_2. \quad (7)$$

Consequently the L^2 length of the normal trajectory before exit is at most $CH(0) = O(\varepsilon^\theta)$.

For completeness, the parabolic estimate used to turn this path-length bound into trapping also survives for $1 < p < 2$. After differentiating the curvature equation m times, its leading integrated term is

$$-\int_{\mathbb{S}^1} \langle A_k \nabla_s^{m+2} k, \nabla_s^{m+2} k \rangle ds.$$

By (4) this controls the highest derivative with constant proportional to $\min\{1, p-1\}$. All remaining coefficients are smooth and bounded on the curvature annulus. One-dimensional interpolation absorbs the intermediate derivatives, giving the same uniform high-Sobolev bounds while the curve remains in the chosen H^4 neighborhood as in the source's parabolic-estimate lemma.

In constant-speed gauge, the change of parametrization is controlled by $\int |k| |G_p| ds$. The uniform curvature bound and (7) therefore give

$$\|\hat{\gamma}(t) - \gamma_*\|_{L^2} \leq C\varepsilon^\theta \quad (0 \leq t < T).$$

Interpolating this small L^2 distance against the uniform high-Sobolev bounds yields

$$\|\hat{\gamma}(t) - \gamma_*\|_{H^4} \leq C\varepsilon^\alpha$$

for some $\alpha > 0$ independent of T . Taking ε small makes the right side strictly smaller than the exit radius, contradicting finite maximality. Hence $T = \infty$.

Equation (7) now gives finite total L^2 path length. The uniform parabolic bounds and interpolation upgrade the resulting L^2 Cauchy property to convergence in every C^m norm. The limit is critical by (6) and continuity of G_p . Finally, the source's compactness argument for the sequence of ambient isometries is independent of the value of p once trapping and finite L^2 path length are known; it identifies a single limiting critical curve for the original flow. This proves part (iii).

Upgrade audit and remaining obstruction

The result was pushed along four further directions.

- **Curvature zeros.** For $1 < p < 2$, the map $z \mapsto |z|^{p-2}z$ is not differentiable at $z = 0$, and its linearization is singular there. Analytic gradient theory and uniform classical parabolicity both fail. A nonsmooth/subgradient theory is required.
- **Global existence and sub-convergence.** The local theorem does not prevent curvature from reaching zero and supplies no global continuation estimate. These are the principal missing inputs for a complete solution of the source’s broad problem.
- **The endpoint $p = 1$.** The longitudinal eigenvalue in (4) becomes zero, and the curvature integrand is nonsmooth. The present proof has a sharp structural endpoint barrier.
- **Quantitative confinement and Huisken’s half-plane conjecture.** The path-length estimate begins only after entry into a local Łojasiewicz–Simon neighborhood and depends on the limiting critical curve. It gives neither an initial-data-only confinement radius nor a half-plane principle.

Bounded title, author, exact-phrase, and arXiv searches through 11 August 2026 found no prior fourth-order subquadratic nonzero-curvature convergence theorem. The related arXiv:1811.06608 concerns a different second-order inextensible flow, while arXiv:2104.03570 treats a modified weak flow for $p \geq 2$. A 2026 survey/thesis still lists the broad subquadratic and half-plane questions as open. Novelty confidence is moderate because this packet extends the source machinery rather than introducing a new global estimate.

References

- [1] M. Pozzetta, *Convergence of elastic flows of curves into manifolds*, arXiv:2007.00582 (2020; revised 2021).
- [2] R. Chill, *On the Łojasiewicz–Simon gradient inequality*, J. Funct. Anal. **201** (2003), 572–601.